PARAMESWARI PALANI

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SUMMARY

Final-year Computer Science undergraduate with hands-on experience building data pipelines and backend systems in Python — including OCR-based extraction, retrieval pipelines, and REST APIs backed by relational and vector databases. Comfortable working across the data lifecycle: extraction, cleaning, transformation, and serving data through APIs. Strong SQL fundamentals and a quick learner of new tooling and infrastructure.

SKILLS

Languages: Python, SQL

Databases: MySQL (relational), ChromaDB / FAISS (vector stores)

Data Pipelines & ETL-style Work: OCR-based data extraction, data cleaning, feature engineering, Retrieval-Augmented Generation (RAG) pipelines, multimodal dataset preparation

Frameworks & APIs: Pandas,Numpy,LangChain, LangGraph, FastAPI, Flask, REST APIs

Applied ML/AI: NLP, LLMs, Vision-Language Models (VLM)

Languages Spoken: English (Fluent)

INTERNSHIPS

Technology Intern — Synergy Maritime Pvt. Ltd., Chennai, TN *June 2025*

OCR & RAG-based Document Intelligence

- Built an OCR-based data extraction pipeline using Python and Tesseract to pull structured text from scanned documents.
- Developed a Retrieval-Augmented Generation (RAG) system with LangChain and FAISS to enable intelligent querying over extracted document data.
- Integrated the Gemini API and Streamlit to deliver a functional, user-facing data assistant prototype.

AI Engineer Intern — Tecnonauts Technology Services Pvt. Ltd., Kakkanad, Kerala (Remote) *June 2025*

Vision-Language Model (VLM)-based System for Crime Scene Reconstruction

- Prepared and structured multimodal (image + text) datasets for model training and analysis.
- Applied deep learning and NLP techniques to process and validate large-scale forensic data.

PROJECTS

NCERT Personalization Engine — AI-Powered Adaptive Learning Platform

- Built backend data infrastructure with FastAPI, LangGraph, ChromaDB, and SQLite to power a personalized learning platform.
- Designed a Retrieval-Augmented Generation (RAG) pipeline and Bayesian Knowledge Tracing (BKT) logic to serve adaptive, context-aware content.
- Developed REST APIs for data-driven chat interactions, quiz evaluation, and user profile management.

AI Women Safety Application — Distress Detection using BERT

- Developed a BERT-based text classification model to detect emergency intent from user messages.
- Built a data collection and alert pipeline via a Flask API, combining sentiment scores and keyword-based risk signals.

Forensic Crime Scene Reconstruction using VLM — Patent No. 202641008176

- Created structured image–caption–QA datasets and optimized training with gradient accumulation and GPU memory handling.
- Built an inference pipeline for direct, structured querying of the trained model.

CERTIFICATES

- Microsoft Certified: Azure AI Fundamentals (04/2026)
- Deloitte: Data Analytics Job Simulation — Certificate of Completion (01/27/2025)
- Kaggle: Pandas Completion (02/06/2025)

EDUCATION

Bachelor of Engineering, Computer Science — St. Joseph's College of Engineering *09/2023 – 05/2027*

Currently pursuing degree • CGPA: 8.87

ACHIEVEMENTS

- Patent Filed – AI-based Forensic Crime Scene Reconstruction using VLM (Patent No: 202641008176)
- MHTECHIN Innovation Challenge 2024 – Hackathon Participant